Adding Aurora Serverless to a Provisioned Aurora Cluster

To Begin with the Lab

Summary of the Lab

In this lab, an existing **Amazon Aurora** provisioned cluster was extended by adding an **Aurora Serverless** reader instance. The new reader was created alongside the existing provisioned writer and reader instances, with its own serverless capacity settings and Availability Zone placement. After the additional reader was added, the cluster still exposed the same writer and reader endpoints, and all read traffic continued through the shared reader endpoint. The lab then demonstrated a failover scenario by changing the failover priority so the serverless instance could take over as the writer. Finally, the failover was executed successfully, leaving the serverless instance as the writer and the provisioned instances as readers.

Understanding Mixed Aurora Clusters with Provisioned and Serverless Instances

A mixed **Amazon Aurora** cluster allows provisioned and serverless instances to work together in the same database cluster. This exists so you can keep a stable provisioned writer while adding serverless capacity for flexible scaling or failover. Aurora uses the same cluster endpoints for reads and writes, even when the underlying instance roles change. By adjusting failover priority, you can control which instance becomes the writer if the primary instance fails or if you trigger a manual failover.

Example:

A production Aurora MySQL cluster runs with one provisioned writer and one provisioned reader. A new Aurora Serverless reader is added to absorb extra read load and act as a failover target. When failover is triggered, the serverless reader becomes the writer and the provisioned instances continue serving reads.

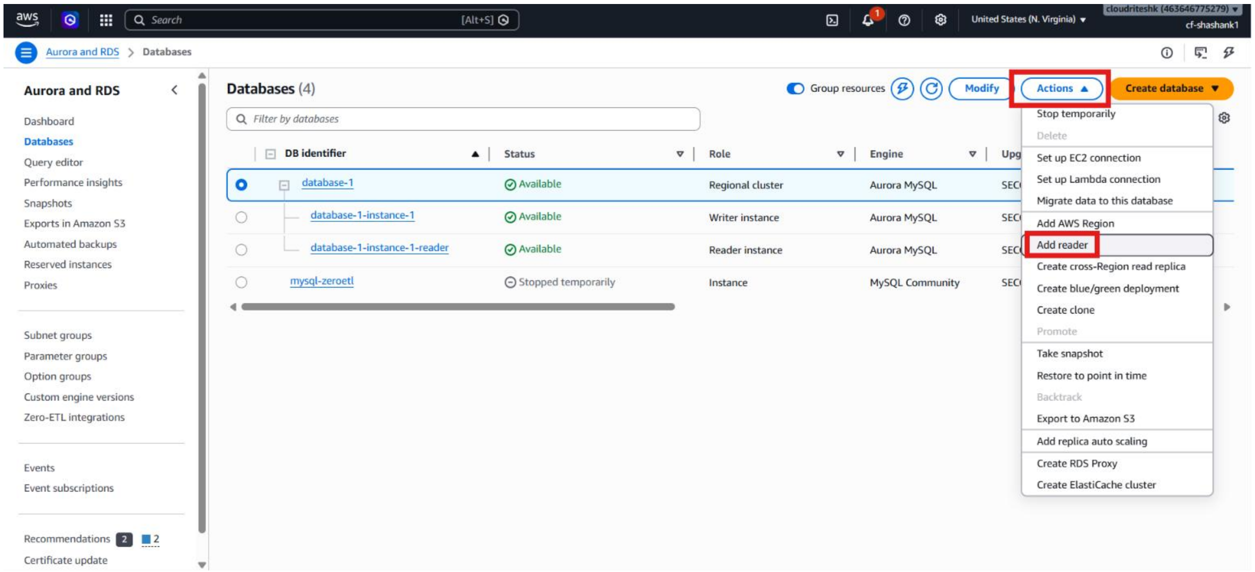
Lab Steps

Step 1 – Open the Existing Aurora Cluster

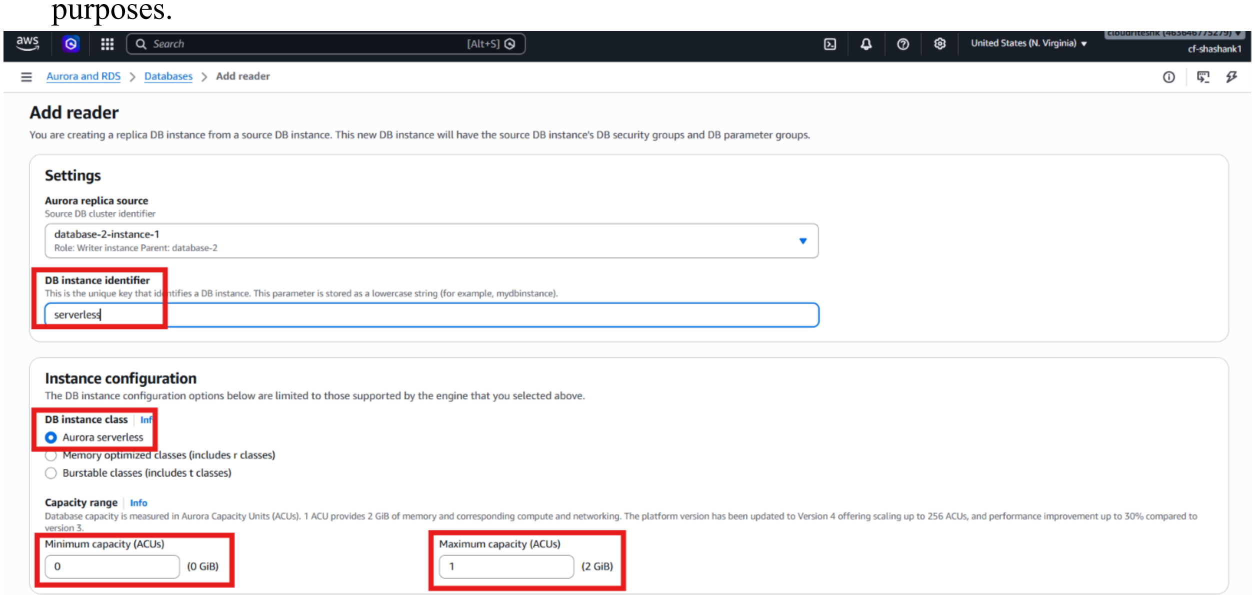
- Sign in to the AWS Management Console.
- Open **Amazon RDS**.
- Select the existing Aurora provisioned cluster from the previous lab.
- Review the current writer and reader instances.
- Confirm that the cluster already has writer and reader endpoints.

Step 2 – Add an Aurora Serverless Reader

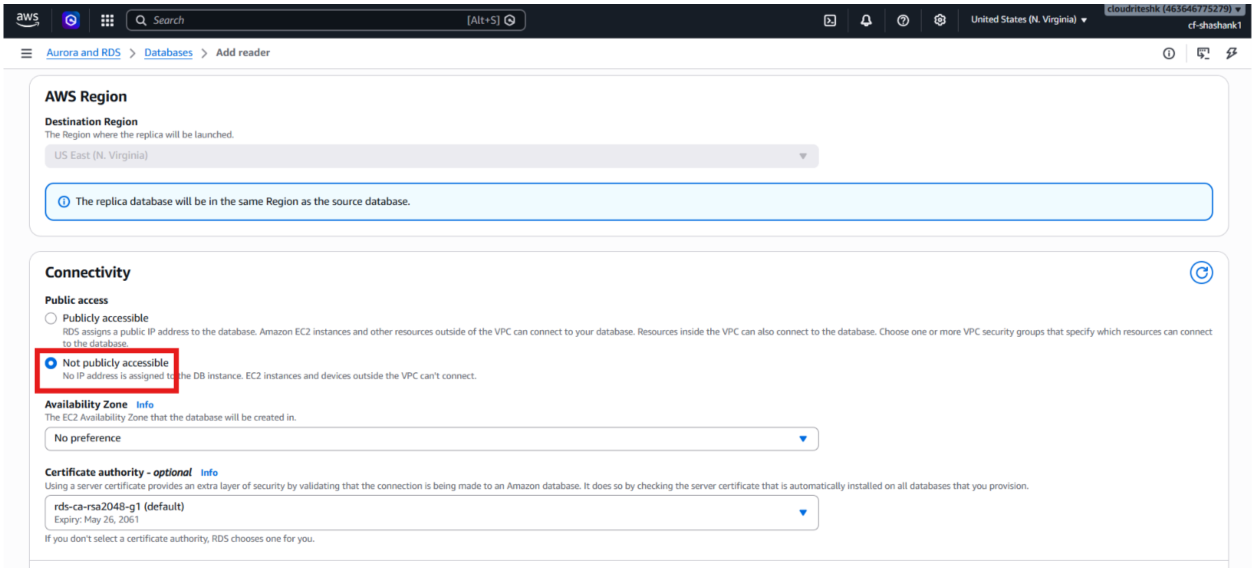
- Click **Actions**.
- Choose **Add reader**.



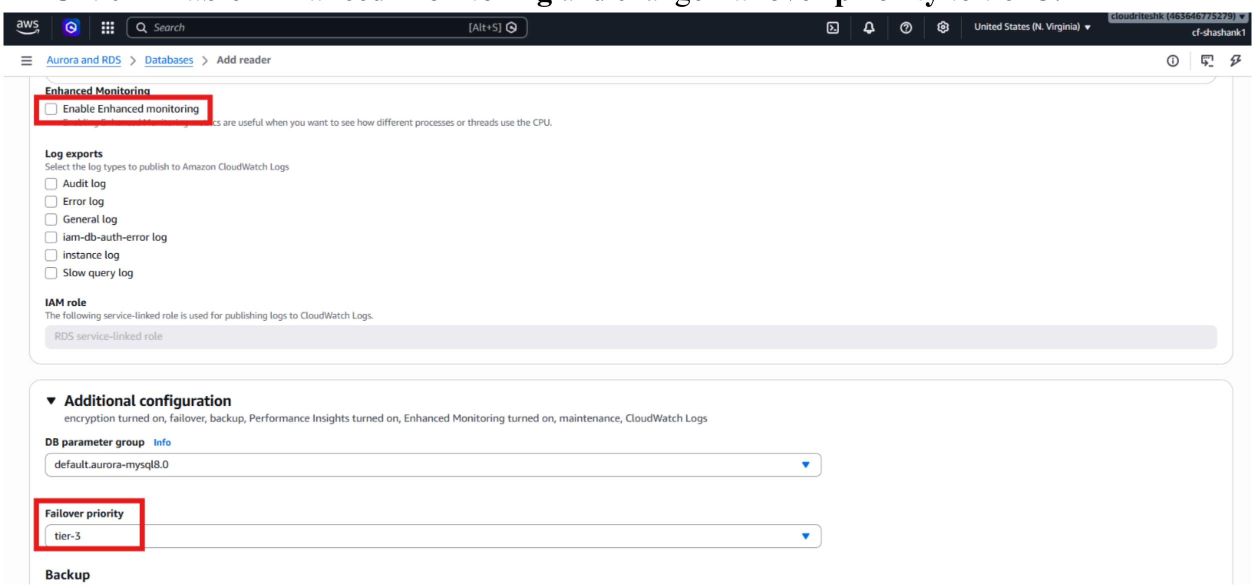
- Enter the new instance identifier: serverless
- Select **Aurora Serverless** as the instance type.
- Configure the minimum and maximum Aurora Capacity Units and ill keep it low for lab purposes.



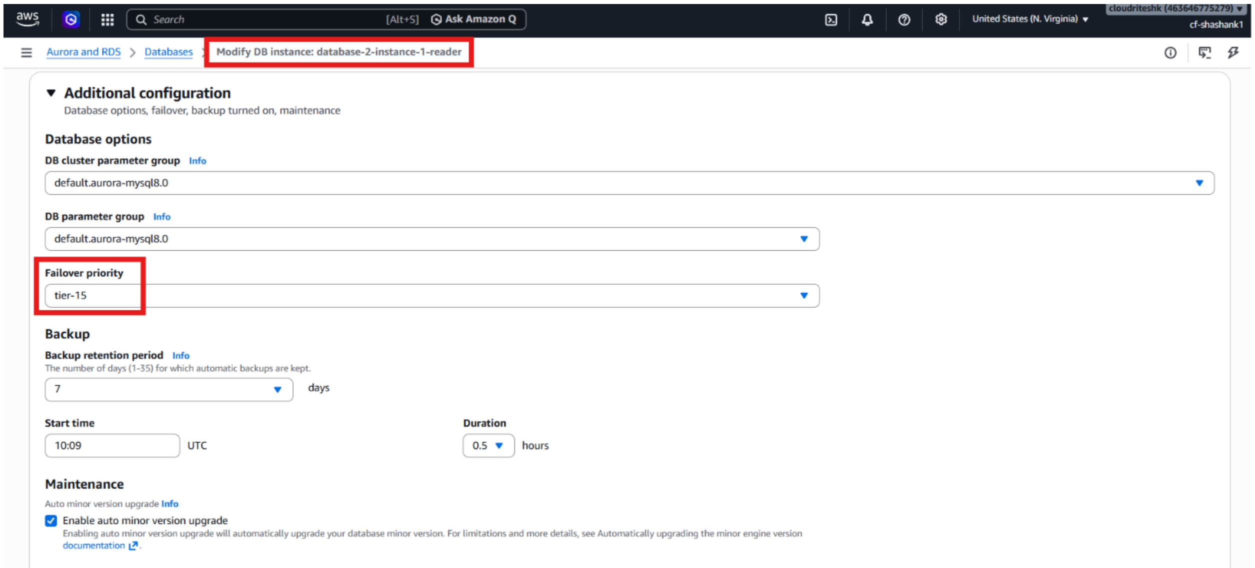
- Keep the instance private and not publicly accessible for lab purposes.



- Leave the defaults as it is.
- Untick **Enable Enhanced Monitoring** and change **Failover priority** to **tier 3**.

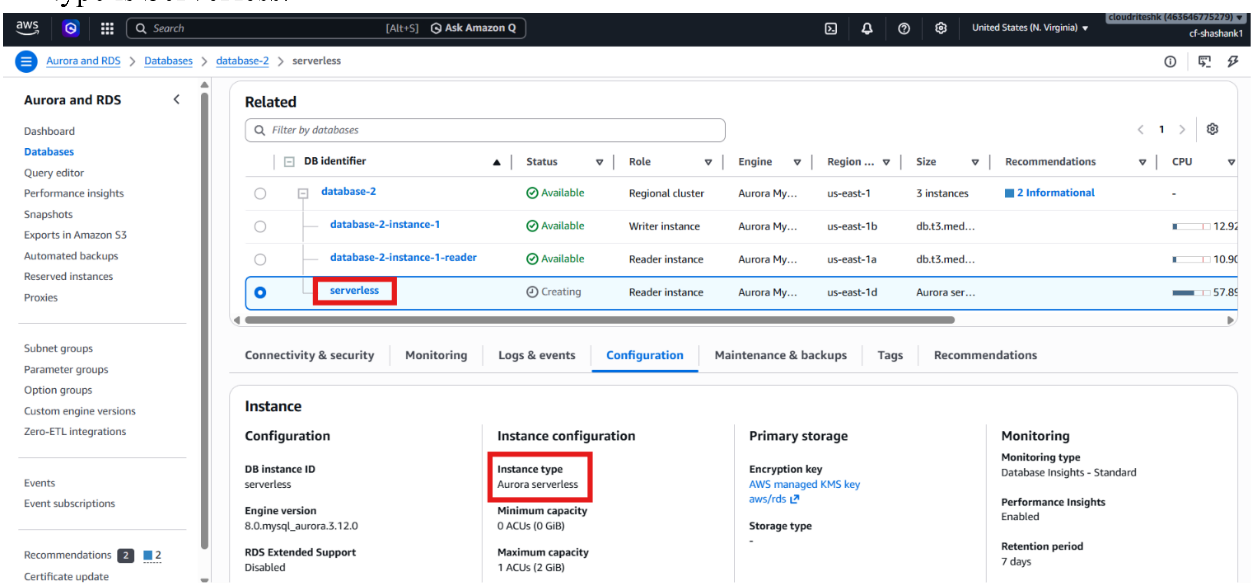


- Click **Add reader**.
- Also modify the **Failover priority** of **reader instance** to **tier 15** for switchover demonstration.



Step 3 – Wait for the Serverless Instance to Become Available

- Wait while Aurora creates the new serverless reader instance.
- Confirm that the instance status becomes **Available** and open **Configuration** tab to verify the type is **Serverless**.

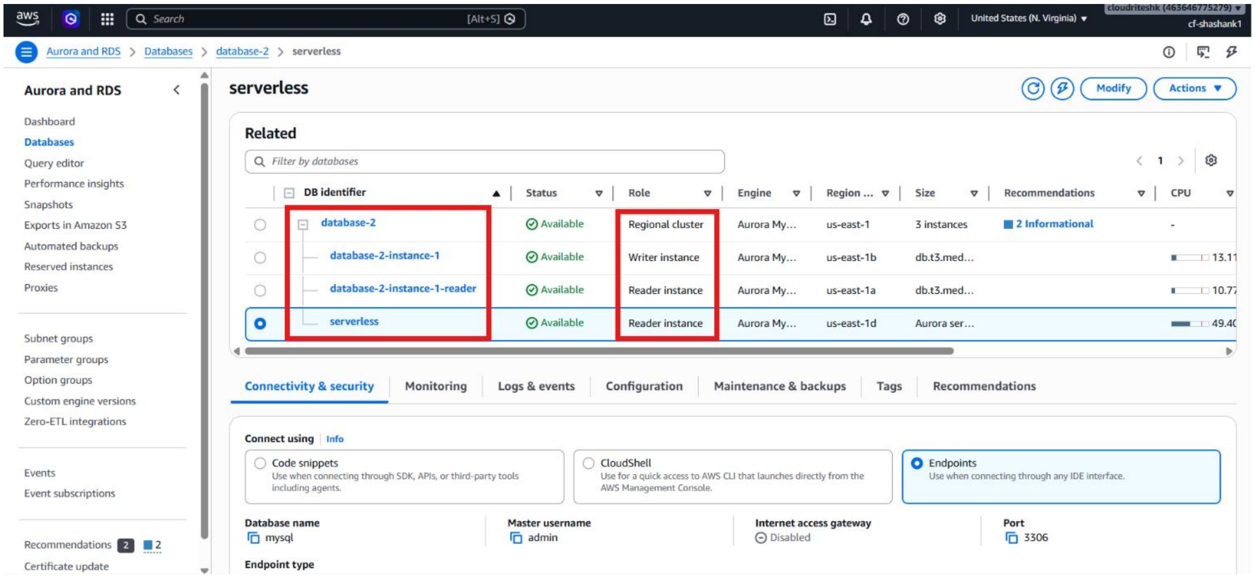


- Confirm that read traffic will be distributed across both reader instances.

Step 4 – Review the Mixed Cluster Setup

- Open the cluster details.
- Confirm that the cluster now contains:
  - One provisioned writer
  - One provisioned reader
  - One Aurora Serverless reader

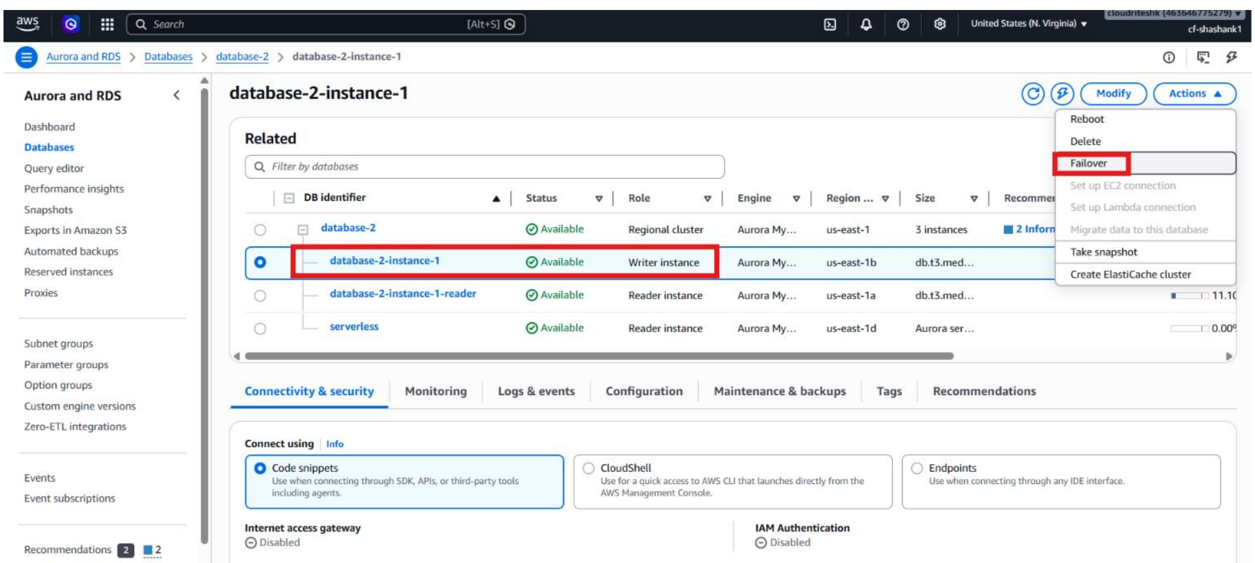
- Verify that the serverless reader is part of the same cluster.



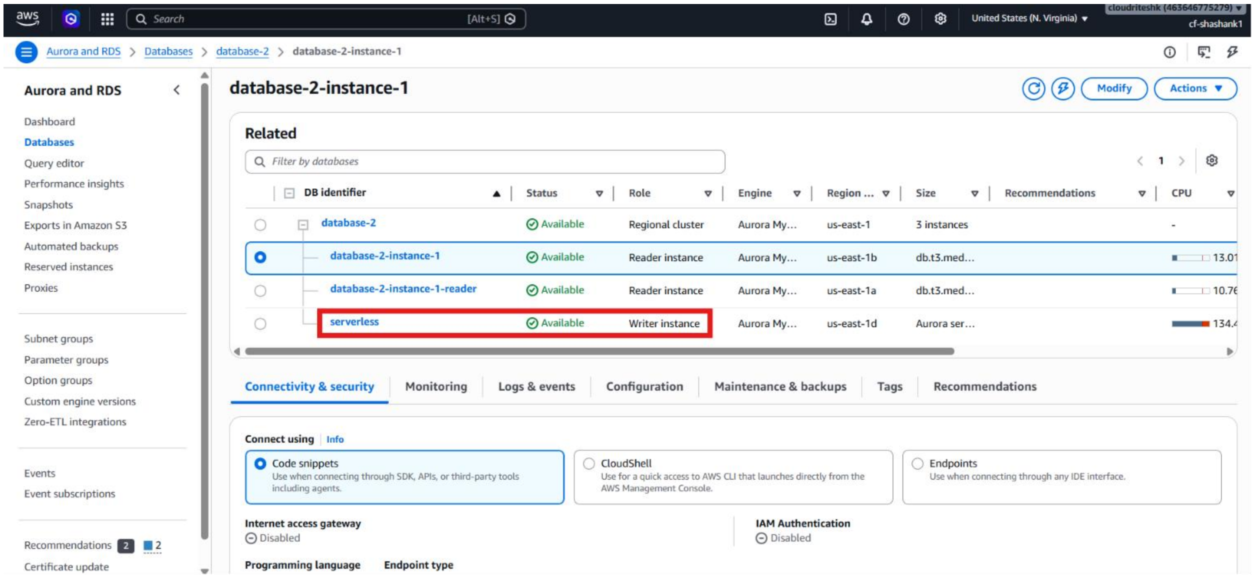
- Confirm that the cluster is now a mixed provisioned and serverless configuration.

Step 5 – Test Failover to the Serverless Instance

- Open the provisioned writer instance.
- Review the failover priority settings.
- Set the serverless instance to a higher failover priority than the provisioned writer if needed.
- Click **Actions** on the cluster.
- Choose **Fail over**.



- Confirm the failover operation.
- Wait for Aurora to complete the switch.
- Verify that the serverless instance becomes the writer.
- Confirm that the provisioned instances become readers.



Verification

- An **Aurora Serverless v2** reader was added successfully to the provisioned cluster.
- The cluster remained accessible through the same writer and reader endpoints.
- The cluster became a mixed provisioned and serverless deployment.
- Failover priority was used to control which instance became writer.
- The failover completed successfully.
- The serverless instance became the writer and the provisioned instances became readers.